

CADSync AI Engineering Report

Part ID: PLATE_A101 | Revision: B

Part Summary

Key	Value
part_id	PLATE_A101
length	240.0
width	128.0
height	15.0
hole_count	10
hole_diameter	10.0
material	Stainless Steel - Satin
tolerance	0.06
revision	B
hole_pattern	[{'Hole_ID': 'H1', 'X': 40, 'Y': 35}, {'Hole_ID': 'H2', 'X': 180, 'Y': 35}, {'Hole_ID': 'H3', 'X': 320, 'Y': 35}, {'Hole_ID': 'H4', 'X': 40, 'Y': 120}, {'Hole_ID': 'H5', 'X': 180, 'Y': 120}, {'Hole_ID': 'H6', 'X': 320, 'Y': 120}, {'Hole_ID': 'H7', 'X': 40, 'Y': 205}, {'Hole_ID': 'H8', 'X': 180, 'Y': 205}, {'Hole_ID': 'H9', 'X': 320, 'Y': 205}, {'Hole_ID': 'H10', 'X': 460, 'Y': 120}]
features	[{'Feature_Type': 'Fillet', 'Value': 3}, {'Feature_Type': 'Extrusion', 'Value': 12}]
configurations	[{'Config': 'Default', 'Export_STEP': True, 'Export_IGES': False, 'Export_DXF': True}]
metadata	{'Designer': 'CADSync AI', 'Project': 'Competition Demo', 'Orientation': 'Landscape'}

Validation Results

Key	Value
Valid	True
Errors	None
Warnings	None

AI Assessment

Key	Value
Anomaly Score	0.0521
Anomaly Flag	False
Manufacturability Risk	0
Risk Probability	5.60%
Estimated Manufacturing Cost	USD 65.97

Optimization Suggestions

Key	Value
Length	240.0
Width	128.0
Hole_Spacing	18.0

- Hybrid optimization executed: GA seed + SciPy refinement.
- Objective prioritizes reduced material usage and manufacturable spacing.

Generated Artifacts

Key	Value
STEP	C:\Users\Tharun B S\Documents\VS Code\CADSync-A\outputs\PLATE_A101\PL
IGES	C:\Users\Tharun B S\Documents\VS Code\CADSync-A\outputs\PLATE_A101\PL
STL	C:\Users\Tharun B S\Documents\VS Code\CADSync-A\outputs\PLATE_A101\PL
DXF	C:\Users\Tharun B S\Documents\VS Code\CADSync-A\outputs\PLATE_A101\PL
Model	C:\Users\Tharun B S\Documents\VS Code\CADSync-A\outputs\PLATE_A101\PL

Engineering Knowledge Graph

CADSync AI Knowledge Graph - PLATE_A101

